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IT469: Human Language Technologies

Fine-Grained Emotion Classification in Tweets:

Investigating Intensifier Effects on Multi-Label Emotion Recognition Using Logistic Regression and BERT

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1. Introduction:

Emotion mining, a subfield of affective computing, involves the detection, analysis, and evaluation of human emotions expressed in various forms, such as text, speech, facial expressions, and movements. Understanding emotions in textual data is particularly significant for numerous real-world applications, including customer feedback analysis, mental health monitoring, and social media sentiment tracking.

In this project, we focus on fine-grained emotion classification within social media texts, particularly tweets. We employ the CBET (Cleaned Balanced Emotional Tweets) dataset, annotated with nine emotion categories: anger, fear, joy, love, sadness, surprise, thankfulness, disgust, and guilt. Our goal is to develop and evaluate models capable of classifying tweets into different emotional categories.

Additionally, this work explores the influence of intensifiers (words like "very" or "extremely") on emotional expression, hypothesizing that intensifiers can modulate the strength or presence of emotions in text. We compare traditional machine learning approaches (Logistic Regression) with transformer-based models (BERT) and assess the impact of incorporating intensifier-aware features.

This project aligns with the growing interest in applying Natural Language Processing (NLP) to understand complex human emotions, improving the granularity and reliability of emotion detection systems.

2. Experiment setup:

In this project, two different methodological approaches were applied to the problem of emotion classification from tweets:

1. **Logistic Regression (LR) with TF-IDF and Intensifier Feature**:
   For the Logistic Regression (LR) approach, we built a structured pipeline rather than feeding raw text directly into the classifier. First, text is transformed via TF-IDF vectorization to capture general word-usage patterns; next, we compare two variants one using only the TF-IDF features and another that augments them with a custom-engineered binary feature flagging the presence of intensifier words (e.g., "very", "extremely", "so"). Both feature sets are combined with FeatureUnion, and the resulting representation is passed to a Logistic Regression classifier. This pipeline design not only leverages broad lexical cues but also explicitly models emotional amplifiers, allowing us to quantify the impact of intensifiers and optimize overall emotion-classification performance.

2. **BERT-based Model with Intensifier Special Tokens**:
   In a deep learning approach, a pre-trained bert-base-uncased model was fine-tuned on the CBET dataset. To explicitly highlight intensifiers to the model, intensifier words were replaced with a special token [INTENSIFIER] during preprocessing. The tokenizer was adapted to recognize this new token, allowing BERT to incorporate intensifier information during the learning process. A classification head was added to the BERT model for predicting emotion categories.

Both approaches aim to improve emotion classification performance by explicitly modeling intensifiers, which are linguistic cues often related to emotional intensity.

2.1 Dataset

The dataset utilized for this project is the **CBET (Cleaned Balanced Emotional Tweets)** dataset. It consists of a collection of tweets labeled with emotional categories. The key characteristics of the dataset are:

Motivation and Task

- **Why was it created?**
  The original motivation was to fill a gap in emotion-mining: existing tweet corpora were either unbalanced across emotion classes or lacked fine-grained emotional annotations. CBET was curated to provide a **balanced** set of examples for anger, fear, joy, love, sadness, surprise, thankfulness, disgust, and guilt, enabling robust training and evaluation of emotion-classification models.

- **Intended use**
  CBET is intended for supervised learning in emotion detection and for comparative benchmarking of different NLP architectures.

- **Funding**
  The public documentation does not specify a dedicated funding source; it appears to have been produced as part of broader academic research initiatives in NLP and sentiment analysis.

Data Collection and Consent

- **Data source**
  Tweets were sampled via the Twitter API from public timelines, filtered by emotion-related keywords.

Corpus Type and Distribution

- **Type**: Short, informal social-media posts (Twitter).

- **Total size**: 81,163 tweets.

- **Training vs. Test split**: 64,930 tweets (80%) for training; 16,233 (20%) for testing

- **Label distribution** (number of tweets per emotion, aggregated across intensifier flag) as shown in Table 1 below:

Table 1: Classification Data Distribution

Emotion	***With intensifier***	***Without intensifier***	***Total***
Anger	*1018*	*8055*	*9073*
Fear	*616*	*8494*	*9110*
Joy	*837*	*10052*	*10889*
Love	*718*	*10922*	*11640*
Sadness	*1430*	*7845*	*9275*
Surprise	*834*	*8453*	*9287*
Thankfulness	*1222*	*7697*	*8919*
Disgust	*1143*	*7504*	*8647*
Guilt	*1140*	*7486*	*8626*
Total	*8661*	*72502*	*81183*

Table 2: Token and Vocabulary Statistics of the Dataset.

Metric	***Count***
Total Tokens	1,230,512
Unique Terms	101,799

- **Data Samples:**

```
----------With Intensifier----------
Random Sample:
                                    text  anger  fear  joy  \
37158  fee light art timcook My Mac is dying because ...    0.0   0.0  0.0
2075   This cop cud ask me if I wud like to sit in hi...    1.0   0.0  0.0
37085  Girls will fuck a man that aint got shit quick...    0.0   0.0  0.0
37952  Kim kar dash ian showed us how FAKE her ass re...    0.0   0.0  0.0
4877               into the darkness wwe #wwe too funny    1.0   0.0  0.0

      love  sadness  surprise  thankfulness  disgust  guilt
37158   0.0      1.0       0.0           0.0      0.0    0.0
2075    0.0      0.0       0.0           0.0      0.0    0.0
37085   0.0      1.0       0.0           0.0      0.0    0.0
37952   0.0      1.0       0.0           0.0      0.0    0.0
4877    0.0      0.0       0.0           0.0      0.0    0.0
```

```
----------Without Intensifier----------
Random Sample:
                                    text  anger  fear  joy  \
30868  KayeMenner JUST FOR YOU #MY #VALENTINE #Card Q...    0.0   0.0  0.0
55793  Give thanks for pets and other animals Donate ...    0.0   0.0  0.0
51594  There further I get through school the more I ...    0.0   0.0  0.0
48325  Another image of our wonderful pisa tower #pis...    0.0   0.0  0.0
10881  life is the art of living with uncertainty wit...    0.0   1.0  0.0

      love  sadness  surprise  thankfulness  disgust  guilt
30868   1.0      0.0       0.0           0.0      0.0    0.0
55793   0.0      0.0       0.0           1.0      0.0    0.0
51594   0.0      0.0       0.0           1.0      0.0    0.0
48325   0.0      0.0       1.0           0.0      0.0    0.0
10881   0.0      0.0       0.0           0.0      0.0    0.0
```

2.2 Methodology

This project focuses on solving a multi-class text classification problem, where the goal is to assign one dominant emotion label to each tweet from a set of predefined emotions: anger, fear, joy, love, sadness, surprise, thankfulness, disgust, and guilt. The project investigates both traditional machine learning methods and transformer-based deep learning models for this task.

Additionally, the study examines the impact of intensifiers (e.g., very, extremely) on emotional expression within tweets by incorporating intensifier-aware features into the classification models.

Approach Overview:

- **Task Type:** Multi-class text classification.
- **Baseline Model:** Logistic Regression with TF-IDF features.
- **Advanced Model:** BERT-based transformer models fine-tuned on the same dataset.
- **Additional experiment:** Incorporation of intensifier features to explore their effect on emotion classification performance.

Model 1: Logistic Regression (Baseline and Intensifier-Aware)

Baseline Logistic Regression:

- **Model Type:** Traditional machine learning classifier for multi-class classification.
- **Feature Engineering:**
    - **TF-IDF Vectorization:** Converts tweets into numerical vectors using unigrams and bigrams (max_features=2000).
- **Classifier: Logistic Regression** model that predicts one dominant emotion label per tweet.

Intensifier-Aware Logistic Regression:

- **Additional Feature:** Introduces a **binary flag** (has_intensifier) indicating the presence of intensifiers in the tweet.
- **Pipeline Design:**
    - Uses **FeatureUnion** to combine:
        - **Textual features:** TF-IDF vectors.

        - **Numeric features:** The intensifier flag (processed via **SimpleImputer** and **StandardScaler**).
- **Classifier:** Same Logistic Regression model, but with the combined feature set to evaluate the impact of intensifiers on prediction.

Model 2: BERT-Based Transformer Models

Two **BERT-based transformer models** were fine-tuned for **multi-class emotion classification**:

Plain BERT Model:

- **Model Type:** Transformer-based deep learning model (BERT).
- **Architecture:** BertForSequenceClassification built on **bert-base-uncased**.
- **Preprocessing:**
    - Tokenization via BertTokenizer with a maximum sequence length of 64 tokens.
- **Training Details:**
    - Epochs: 2
    - Batch size: 4
    - Learning rate and optimization handled via Hugging Face's Trainer and TrainingArguments.

Intensifier-Aware BERT Model:

- **Enhancement:** Prepends a **custom token** ([INTENSIFIER_1] or [INTENSIFIER_0]) to the input sequence, indicating whether an intensifier is present in the tweet.
- **Purpose:** Allows the model to incorporate intensifier cues directly into its input representation, potentially improving its sensitivity to emotionally amplified language.

3. Evaluation and results

In this section, we evaluate the performance of the different models based on **recall**, **precision**, and **macro F1-score**.

Table 3: Evaluation Metrics for Baseline and Intensifier-Aware Models

MODEL	RECALL	PERCISION	MACRO F1
LR (TEXT)	**0.73**	**0.41**	**0.52**
LR (TEXT + INTENSIFIER)	**0.73**	**0.41**	**0.52**
BERT (TEXT)	**0.8677**	**0.8687**	**0.8678**
BERT (TEXT + INTENSIFIER)	**0.8575**	**0.8583**	**0.8575**

4. Discussion

The best-performing model is **BERT (Text Only)**, achieving the highest recall (0.8654), precision (0.8687), and macro F1-score (0.8678).

The BERT model's success is mainly attributed to its ability to capture deep contextual information and semantic nuances within the text, without the need for additional handcrafted features like the intensifier flag. The minimal performance gap between **BERT (Text Only)** and **BERT (Text + Intensifier)** suggests that BERT already understands the effect of intensifiers inherently through its pre-trained language representations.

The figure below shows examples for each of the nine emotion categories. These examples were used to test the model and illustrate that **BERT (Text Only)** was able to correctly classify all of them:

Cases

```
[ ] emotion_labels = ['anger', 'fear', 'joy', 'love', 'sadness', 'surprise', 'thankfulness', 'disgust', 'guilt']
```

Figure 4.1: Defined Emotion Labels for Classification Task

Anger

```
[ ] # Predict
    text = "I'm furious about how they treated me!"
    predicted_class_idx = predict_emotion(text, model_text, tokenizer)

    # predicted emotion
    print(f"Predicted Emotion: {emotion_labels[predicted_class_idx]}")
```

```
Predicted Emotion: anger
```

Figure 4.2: Example Prediction for the 'Anger' Emotion Class

Fear

```
[ ] # Predict
    text = "Is there someone behind me?"
    predicted_class_idx = predict_emotion(text, model_text, tokenizer)

    # predicted emotion
    print(f"Predicted Emotion: {emotion_labels[predicted_class_idx]}")
```

```
Predicted Emotion: fear
```

Figure 4.3: Example Prediction for the 'Fear' Emotion Class

Joy

```
[ ] # Predict
    text = "OMG! LIVERPOOL JUST WON THE LEAGUE"
    predicted_class_idx = predict_emotion(text, model_text, tokenizer)

    # predicted emotion
    print(f"Predicted Emotion: {emotion_labels[predicted_class_idx]}")
```

```
Predicted Emotion: joy
```

Figure 4.4: Example Prediction for the 'Joy' Emotion Class

Love

```
[ ] # Predict
    text = "I will never leave you"
    predicted_class_idx = predict_emotion(text, model_text, tokenizer)

    # predicted emotion
    print(f"Predicted Emotion: {emotion_labels[predicted_class_idx]}")
```

```
Predicted Emotion: love
```

Figure 4.5: Example Prediction for the 'Love' Emotion Class

Thankfulness

```
[ ] # Predict
    text = "I don't know what would i do without you"

    predicted_class_idx = predict_emotion(text, model_text, tokenizer)

    # predicted emotion
    print(f"Predicted Emotion: {emotion_labels[predicted_class_idx]}")
```

```
Predicted Emotion: thankfulness
```

Figure 4.6: Example Prediction for the 'Thankfulness' Emotion Class

Surprise

```
[ ] # Predict
    text = "I can not believe my eyes!"

    predicted_class_idx = predict_emotion(text, model_text, tokenizer)

    # predicted emotion
    print(f"Predicted Emotion: {emotion_labels[predicted_class_idx]}")
```

```
Predicted Emotion: surprise
```

Figure 4.7: Example Prediction for the 'Surprise' Emotion Class

Sadness

```
[ ] # Predict
    text = "I am devastated about my results!"
    predicted_class_idx = predict_emotion(text, model_text, tokenizer)

    # predicted emotion
    print(f"Predicted Emotion: {emotion_labels[predicted_class_idx]}")
```

```
Predicted Emotion: sadness
```

```
[ ] # Predict
    text = "I feel so empty and heartbroken right now."
    predicted_class_idx = predict_emotion(text, model_text, tokenizer)

    # predicted emotion
    print(f"Predicted Emotion: {emotion_labels[predicted_class_idx]}")
```

```
Predicted Emotion: sadness
```

Figure 4.8: Example Prediction for the 'Sadness' Emotion Class

Disguest

```
[ ] # Predict
    text = "Ew! you just ate that?"

    predicted_class_idx = predict_emotion(text, model_text, tokenizer)

    # predicted emotion
    print(f"Predicted Emotion: {emotion_labels[predicted_class_idx]}")

    Predicted Emotion: disgust
```

Figure 4.9: Example Prediction for the 'Disguest' Emotion Class

Guilt

```
[ ] # Predict
    text = "Are you mad?"

    predicted_class_idx = predict_emotion(text, model_text, tokenizer)

    # predicted emotion
    print(f"Predicted Emotion: {emotion_labels[predicted_class_idx]}")

    Predicted Emotion: guilt
```

Figure 4.10: Example Prediction for the 'Guilt' Emotion Class

5. Conclusion

In this project, we focused on evaluating emotion classification models based on raw text and intensifier information. Our main finding is that **BERT (Text Only)** outperforms all other models, achieving the highest recall, precision, and macro F1-score without the need for additional intensifier features. This demonstrates BERT's strong ability to understand emotional context natively.